Lab Day: Deploy + Capstone Debug

Phase: Backend — Django
Prerequisites: Django `05_Security_and_Deployment.md`; a CRUD+auth app from Lab `06` or CampusCart MVP slice
Goal: Production-like config verified + at least one **live or staging URL** (or documented blocked-by-platform note with local `check --deploy` green)
Previous module: Lab CRUD Auth Consolidation
Next module: Flask Overview

Learning Outcomes

- Apply `load_dotenv()`, `DEBUG=False` simulation, and WhiteNoise/`STORAGES` from Module 05
- Run and interpret `python manage.py check --deploy`
- Debug static files, ALLOWED_HOSTS, and database connection failures calmly
- Update README with deploy steps and env var list

Must-pass checklist

Done when

All boxes below must be checked before you submit the deploy tag.

- [ ] `.env` gitignored; `.env.example` committed
- [ ] `load_dotenv(BASE_DIR / ".env")` present in settings
- [ ] `STORAGES["staticfiles"]` configured (Django 5.x pattern)
- [ ] `python manage.py check --deploy` run with `DJANGO_DEBUG=False`
- [ ] `collectstatic` succeeds
- [ ] Gunicorn smoke test **or** platform deploy started
- [ ] README: env vars table + demo accounts (non-production passwords)
- [ ] Screenshots of live/staging **or** local production-mode proof

Success is judged on **env hygiene +** **`check --deploy`**, not only a public URL.

Where used? Staging/live CampusCart or student portal demos — same checklist recruiters expect on GitHub READMEs.

Common Error: Deploying with `DEBUG=True` or empty `ALLOWED_HOSTS` → `DisallowedHost` or leaked tracebacks. Set env on the platform first.

Debug playbook

Problem	First checks
DisallowedHost	ALLOWED_HOSTS includes platform hostname
CSS missing on deploy	WhiteNoise middleware order; collectstatic in build
DB connection refused	Host/user/password; PaaS IP allowlist
500 with DEBUG=False	Check logs; temporary logging; never turn DEBUG=True on public URL long-term
Secret key warning	Generate new key; set env; rotate if leaked

Skill-level rubric (today)

Level	Evidence
Not started	No check --deploy attempt
Guided	Passes checks only with help
Independent	check --deploy clean + collectstatic + README env table
Interview-ready	Explains why DEBUG=False and env secrets matter in interviews

Assignment

Submit deploy checklist filled + repo tag `lab07-deploy`. Capstone M4 advanced if URL live.